MATH INQUIRERS HOLIDAY PROGRAM

Question Bank

Core, Think Harder and Challenge questions for every lesson, with answers

Notice it. Test it. Explain why.

Question Bank • Ages 9-11 (finishing P4)

Math Inquirers P4 Ages 9-11. Notice it. Test it. Explain why.

How to use this bank

- Each lesson has three levels. Every child starts with Core, then moves up when they are ready. Challenge questions are the ones to talk about together.
- CORE builds fluency with the main idea of the lesson. Use it for warm-ups and for children who need confidence.
- THINK HARDER goes a step further: two-step problems, patterns with a hidden rule, unknowns and larger numbers.
- CHALLENGE questions are non-routine. They ask children to find every way, work backwards, test a case, justify an answer or notice what is missing. This is the style of thinking used in gifted-education-type problems. It is not an official GEP standard, and this program is not affiliated with MOE.
- Ask for the reason every time: "How do you know?" A correct answer with no reason is only half done.
- Answers are at the back. Do not hand the answer pages to children.

What each lesson practises

Lesson 1	Factor Hunters (Factors and multiples)	6 + 8 + 9 questions
Lesson 2	Big Number Puzzles (Numbers to 100 000)	6 + 8 + 9 questions
Lesson 3	Operation Detectives (Four operations)	6 + 9 + 9 questions
Lesson 4	Fraction Frontier (Fractions)	6 + 9 + 9 questions
Lesson 5	Decimal Detectives (Decimals)	6 + 8 + 9 questions
Lesson 6	Angle Turns (Angles)	6 + 9 + 9 questions
Lesson 7	Area Architects II (Area and Perimeter)	6 + 8 + 9 questions
Lesson 8	Symmetry and Nets (Symmetry and Nets)	6 + 8 + 9 questions
Lesson 9	Data Detectives (Tables and line graphs)	6 + 8 + 9 questions
Lesson 10	Challenge Day (Mixed heuristics)	6 + 8 + 9 questions

Lesson 1: Factor Hunters

Factors and multiples • Week 1

CORE

C1. Write all the factors of 12.

C2. Write the first five multiples of 7.

C3. Is 4 a factor of 52? How do you know?

C4. Find a common multiple of 3 and 5 that is less than 20.

C5. Write all the factor pairs of 20.

C6. What is the smallest number that is a multiple of both 4 and 6?

THINK HARDER

T1. How many factors does 72 have? List them in order.

T2. One lighthouse flashes every 6 seconds. Another flashes every 9 seconds. They flash together now. After how many seconds will they next flash together?

T3. Find all the common factors of 24 and 36.

T4. I am a multiple of 7 between 50 and 80. My digits add up to 11. Who am I?

T5. Ben shares 28 stickers equally among some friends with none left over. He has more than 1 friend and gives each friend more than 1 sticker. How many friends could he have? Find all the answers.

T6. Which number less than 30 has the most factors?

T7. Which of these numbers are multiples of both 4 and 9: 54, 72, 96, 108?

T8. Mrs Tan has 36 pencils and 48 erasers. She puts them into identical packs with nothing left over. What is the greatest number of packs she can make? What is in each pack?

CHALLENGE (non-routine: justify, find every way)

X1. How many numbers from 1 to 100 are multiples of 4 or multiples of 6 (or both)?

X2. Mei has between 30 and 50 eggs. When she packs them in 4s, 1 egg is left. When she packs them in 5s, 1 egg is left. How many eggs does she have?

X3. What is the smallest number that has exactly 6 factors?

X4. Ali counts in 6s from 6. Mei counts in 8s from 8. Both stop at 100. How many numbers do they both say?

X5. Ravi runs one lap of a track every 4 minutes. Siti runs one lap every 6 minutes. They start together at the start line. How many laps has each run when they are next at the start line together?

X6. Find a number between 20 and 40 that has exactly 4 factors and is odd.

X7. Three school bells ring every 4 minutes, every 6 minutes and every 10 minutes. They all ring together at 9:00 am. When do they next all ring together?

X8. I am a number less than 100. When I am divided by 2, 3, 4, 5 or 6, there is always 1 left over. I am bigger than 1. Who am I?

X9. There are 20 closed lockers, numbered 1 to 20. Child 1 opens every locker. Child 2 changes every 2nd locker (opens it if it is closed, closes it if it is open). Child 3 changes every 3rd locker, and so on, up to child 20. Which lockers are open at the end?

Lesson 2: Big Number Puzzles

Numbers to 100 000 • Week 1

CORE

C1. In 64 205, what is the digit 4 worth?

C2. Write fifty-three thousand and eighty in numerals.

C3. Round 27 649 to the nearest thousand.

C4. What number is 1000 less than 70 230?

C5. Put these numbers in order, smallest first: 45 099, 45 909, 45 990, 45 900.

C6. Round 6352 to the nearest hundred.

THINK HARDER

T1. I am a 5-digit number. All my digits are the same. Rounded to the nearest ten thousand, I am 60 000. Who am I?

T2. Use the digit cards 1, 4, 6, 7 and 9 once each. Make the 5-digit number that is closest to 70 000.

T3. Find the rule and write the next two numbers: 17 250, 19 500, 21 750, ____, ____

T4. How many whole numbers round to 3500 when rounded to the nearest hundred?

T5. What is the smallest whole number that rounds to 80 000 when rounded to the nearest thousand?

T6. Estimate 8 × 6102 by rounding 6102 to the nearest thousand first. Is the exact answer more or less than your estimate?

T7. How many 5-digit numbers have all five digits the same?

T8. What is the largest 5-digit odd number with all its digits different?

CHALLENGE (non-routine: justify, find every way)

X1. Ali counts on in 1000s, starting at 3456. He stops at the last number below 100 000. What is the last number he says? How many numbers does he say?

X2. How many whole numbers round to 45 000 to the nearest thousand AND to 50 000 to the nearest ten thousand?

X3. A palindrome reads the same forwards and backwards, like 12 321. How many 5-digit palindromes are there between 40 000 and 50 000?

X4. Mei rounds a 4-digit number to the nearest thousand and multiplies by 6. She gets 24 000. What are the smallest and largest numbers she could have started with?

X5. Find all the 5-digit palindromes that round to 50 000 when rounded to the nearest thousand.

X6. Ben writes all the numbers from 99 990 to 100 000. How many digits does he write altogether?

X7. How many 5-digit numbers have digits that add up to 3?

X8. A 5-digit number has all its digits different, and its digits add up to 10. What is the largest it can be? What is the smallest?

X9. The distance counter on Mr Tan's car shows 39 993 km. This number is a palindrome: it reads the same forwards and backwards. What is the next palindrome the counter will show? How many more kilometres is that?

Lesson 3: Operation Detectives

Four operations • Week 2

CORE

C1. Work out 18 + 6 × 3.

C2. Work out (18 + 6) × 3.

C3. Work out 56 ÷ 7 × 2.

C4. Work out 90 – 30 ÷ 5.

C5. Work out 2 345 × 3.

C6. Work out 4 236 ÷ 6.

THINK HARDER

T1. There are 2 340 red and blue beads. There are 4 times as many blue beads as red beads. How many red beads are there?

T2. Put ONE pair of brackets in 3 + 5 × 6 – 2 to make the answer 46.

T3. Write +, –, × or ÷ in each gap to make this true: 12 __ 4 __ 5 = 15

T4. Think of a number. Multiply it by 5. Add 35. Divide by 5. Take away the number you started with. The answer is always 7. Explain why it always works. A bar model can help.

T5. Work out 120 – 4 × (18 + 7).

T6. In this chain, each number after the first two is the sum of the two before it. Fill in the gaps: 3, __, __, 17, __

T7. A plate of noodles costs $4 and a drink costs $2. Ravi buys 3 plates of noodles and 2 drinks. Write ONE number sentence for the total cost and work it out.

T8. Is 50 – 20 – 10 the same as 50 – (20 – 10)? Explain.

T9. I think of a number. I multiply it by 8 and then take away 15. The answer is 425. What is my number?

CHALLENGE (non-routine: justify, find every way)

X1. Use four 4s and any of +, –, ×, ÷ and brackets to make 5.

X2. Start at 7. If a number is even, halve it. If it is odd, multiply it by 3 and add 1. How many steps does it take to reach 1?

X3. Two numbers add up to 1 000. One number is 4 times the other. What are the two numbers?

X4. Ali has $56 more than Ben. Together they have $300. How much money does Ben have?

X5. Use 3, 4, 5 and 6 once each, with any signs and brackets, to make 24. Find two different ways.

X6. The same number goes in every box: □ × □ – □ = 30. What is the number?

X7. Put + or – in each gap so that the answer is 7: 1 __ 2 __ 3 __ 4 __ 5 __ 6 = 7. Find ALL the ways.

X8. Mei spent half of her money on a book. Then she spent $6 on lunch. Then she spent half of what was left on a gift. She had $9 left. How much money did she have at first?

X9. In a number chain, each number after the first two is the sum of the two before it. The 5th number is 30 and the 6th number is 49. What is the 1st number?

Lesson 4: Fraction Frontier

Fractions • Week 2

CORE

C1. Write 11/3 as a mixed number.

C2. Write 2 3/4 as an improper fraction.

C3. Find 1/4 of 32.

C4. Find 2/3 of 18.

C5. Which is bigger, 5/6 or 2/3?

C6. Work out 3/10 + 2/5.

THINK HARDER

T1. Ben ate 7/4 pizzas and Ravi ate 1 5/8 pizzas. Who ate more?

T2. Siti has 40 beads. 3/8 of them are blue. How many beads are NOT blue?

T3. Put these fractions in order, smallest first: 2/3, 1/2, 5/6, 7/12

T4. Who am I? I am an improper fraction. My denominator is 6. I am between 2 and 3. My numerator is an odd number. I am closer to 3 than to 2. Make a list, then cross out the ones that do not fit.

T5. Find ALL the fractions with denominator 10 that are bigger than 1/5 but smaller than 1/2.

T6. A ribbon is 3 m long. Mei cuts it into pieces that are each 1/4 m long. How many pieces does she get?

T7. How many quarters are there in 4 3/4?

T8. Ravi says 3/7 is bigger than 3/5 because 7 is bigger than 5. Is he right? Explain.

T9. 2/5 of a number is 16. What is the number?

CHALLENGE (non-routine: justify, find every way)

X1. How many fractions with denominator 12 are bigger than 1/4 but smaller than 2/3?

X2. Mei spent 1/3 of her money on a book and 1/4 of her money on a pen. She had $25 left. How much money did she have at first?

X3. A tank is 3/8 full of water. After 15 litres more are poured in, it is 3/4 full. How many litres does the full tank hold?

X4. Use two of the digit cards 2, 3, 5 and 7 to make a fraction smaller than 1 that is as close to 1 as possible.

X5. In a bag of marbles, 1/4 are red and 1/3 are blue. The other 10 marbles are green. How many marbles are in the bag?

X6. A cake is cut into 12 equal pieces. Mei eats 1/4 of the cake and Ben eats 1/3 of the cake. How many pieces are left?

X7. I am a fraction equal to 3/4. My numerator and denominator add up to 28. Who am I?

X8. Half of a number is 3 more than a third of the number. What is the number?

X9. Siti read 2/5 of her book on Monday. On Tuesday she read half of the pages that were left. She still has 45 pages to read. How many pages are in the book?

Lesson 5: Decimal Detectives

Decimals • Week 3

CORE

C1. Write 3 ones, 4 tenths and 7 thousandths as a decimal.

C2. What is the value of the digit 6 in 12.365?

C3. Which is greater, 0.9 or 0.89? How do you know?

C4. Round 7.46 to 1 decimal place.

C5. Round 15.96 to 1 decimal place.

C6. Ali buys 3 pens at $1.25 each. How much does he pay?

THINK HARDER

T1. Find ALL the digits that can go in the box: 3.□5 < 3.42

T2. A number with 2 decimal places rounds to 10 when rounded to the nearest whole number. What is the smallest it can be? The largest?

T3. What number is halfway between 2.6 and 2.7? What number is halfway between 2.6 and 2.61?

T4. A ribbon 4.8 m long is cut into 6 equal pieces. How long is each piece? Ravi uses 4 of the pieces. How much ribbon does he use?

T5. Siti buys 4 notebooks that cost the same. She pays with a $10 note and gets $2.60 change. How much does one notebook cost?

T6. Write these in order, smallest first: 0.6, 0.066, 0.606, 0.66, 0.06

T7. Mei is 1.38 m tall. Ben is 0.15 m taller than Mei. Ali is 0.09 m shorter than Ben. How tall is Ali?

T8. Use the digit cards 2, 4, 7 and a decimal point once each (the point goes between two digits). Which number is closest to 50?

CHALLENGE (non-routine: justify, find every way)

X1. A number with 3 decimal places rounds to 2.46 when rounded to 2 decimal places. Its digits add up to 20. What is the number?

X2. 3 apples and 1 pear weigh 0.95 kg. 1 apple and 1 pear weigh 0.45 kg. All the apples weigh the same. What is the mass of the pear?

X3. Ravi has 10 coins. They are all 20¢ and 50¢ coins. Altogether they make $3.80. How many of each coin does he have?

X4. Ben rounds a number with 2 decimal places to the nearest whole number and gets 6. Mei rounds the same number to 1 decimal place and gets 5.5. Find ALL the numbers it could be.

X5. A pen costs $0.85 more than a pencil. 2 pens and 1 pencil cost $4.70 altogether. How much does a pencil cost?

X6. How many numbers with exactly 2 decimal places are between 3.9 and 4.1? (Do not count 3.9 and 4.1.)

X7. Use the digit cards 1, 2, 4, 8 and a decimal point once each to make a number with 2 decimal places. Which number is closest to 50?

X8. Ali, Ben and Siti share $9.20. Ben gets $0.50 more than Ali. Siti gets $1.30 more than Ben. How much does each child get?

X9. A jug full of water weighs 1.45 kg. When it is half full, it weighs 0.85 kg. What is the mass of the empty jug?

Lesson 6: Angle Turns

Angles • Week 3

CORE

C1. How many degrees are there in a half turn?

C2. How many degrees are there in 3/4 of a turn?

C3. Ali faces E. He turns 90° clockwise. Which direction does he face now?

C4. Mei faces N. She turns 135° anticlockwise. Which direction does she face now?

C5. What is the angle between the hands of a clock at 6 o'clock?

C6. A line from a corner of a rectangle cuts the 90° corner into two angles. One angle is 64°. What is the other?

THINK HARDER

T1. Siti faces SE. She turns 225° clockwise. Which direction does she face now?

T2. What is the angle between the clock hands at 10 o'clock? At 2 o'clock? What do you notice?

T3. How many degrees does the minute hand turn from 7:15 to 7:50?

T4. A robot starts facing N. Ben wants it to turn from N to W using two DIFFERENT cards from: 45° clockwise, 90° clockwise, 135° anticlockwise, 180°. Find ALL the pairs that work.

T5. Mei faces W. She turns clockwise until she faces SE. How many degrees does she turn?

T6. Four rays come out of one point, making 3 angles side by side: 25°, 40° and 25°. How many angles are there altogether? What is the biggest?

T7. Ravi faces N and turns anticlockwise to face S. Then he turns anticlockwise again to face E. How many degrees has he turned altogether?

T8. In a rectangle, a line from one corner makes a 27° angle with the long side. What angle does it make with the short side?

T9. The corner of a square is cut into two angles. One angle is 4 times as big as the other. Find both angles.

CHALLENGE (non-routine: justify, find every way)

X1. What is the angle between the hands of a clock at 3:30?

X2. Five rays come out of one point and make 4 equal angles side by side. The angle between the two outside rays is 120°. How many angles can you find altogether? How many of them are right angles?

X3. A robot faces N. It turns 45° clockwise, again and again. After how many turns does it first face N again? Which way does it face after 30 turns?

X4. The minute hand turns 90°. How many minutes have passed? The hour hand turns 90°. How many hours have passed?

X5. How many degrees does the hour hand turn in 20 minutes?

X6. Mei faces S and turns clockwise to face NE. Ali faces S and turns anticlockwise to face NE. How many more degrees did Mei turn than Ali?

X7. What is the angle between the hands of a clock at 9:30?

X8. The corner of a square is cut into 3 angles. The middle angle is twice the smallest. The biggest is 3 times the smallest. Find all three angles.

X9. Two squares share one corner O and overlap, like Q5(c). The angle covered by both squares together (from the outside side of one square to the outside side of the other) is 160°. What is the angle where the squares overlap?

Lesson 7: Area Architects II

Area and Perimeter • Week 4

CORE

C1. Find the area of a rectangle 9 cm long and 7 cm wide.

C2. Find the perimeter of a rectangle 15 cm long and 4 cm wide.

C3. A square has a perimeter of 36 cm. Find its area.

C4. A rectangle has an area of 56 cm². Its length is 8 cm. Find its perimeter.

C5. A figure is made of a 6 cm by 4 cm rectangle and a 3 cm by 2 cm rectangle joined together without overlapping. Find its area.

C6. A 4 cm by 4 cm square is cut from one corner of a 10 cm by 10 cm square. Find the area left. Has the perimeter changed?

THINK HARDER

T1. List every rectangle with whole-cm sides and an area of 30 cm². Which one has the smallest perimeter?

T2. A rectangle has whole-cm sides and a perimeter of 20 cm. List all the possible rectangles and their areas. Which has the largest area?

T3. A 3 cm by 3 cm square is cut from each of the four corners of a 12 cm by 8 cm rectangle. Find the area and the perimeter of the shape that is left.

T4. Three squares with sides 2 cm, 3 cm and 4 cm stand side by side in a row on the same line, smallest on the left. Find the area and the perimeter of the figure.

T5. A path 2 m wide goes all round a lawn 10 m long and 6 m wide. Find the area of the path.

T6. Ali joins 8 square tiles, each 1 cm by 1 cm, edge to edge. What is the smallest perimeter his shape can have? The largest?

T7. A 6 cm by 4 cm rectangle is cut into two 6 cm by 2 cm rectangles, which are put end to end to make a 12 cm by 2 cm rectangle. How much longer is the new perimeter?

T8. A rectangle has whole-cm sides, an area of 40 cm² and a perimeter of 26 cm. Find its length and breadth.

CHALLENGE (non-routine: justify, find every way)

X1. A square is cut into two identical rectangles. Each rectangle has a perimeter of 24 cm. Find the area of the square.

X2. A gap is cut into the middle of the top side of a 10 cm by 6 cm rectangle. The gap is 2 cm wide and 4 cm deep, with square corners. Find the area and the perimeter of the shape that is left.

X3. Ravi has 24 m of fencing to make a rectangular pen with whole-metre sides. What is the largest area he can fence?

X4. Ravi uses the same 24 m of fencing, but now one side of the pen is a wall and needs no fence. What is the largest area now?

X5. Two 8 cm by 2 cm rectangles cross at their middles to make a plus shape. Find the area and the perimeter of the plus shape.

X6. Siti's garden is a square with an area of 49 m². She makes it 2 m longer and 2 m wider. How much more area does it have now?

X7. Mei makes a rectangle with 36 square tiles (each 1 cm by 1 cm). Ben makes a different rectangle with 36 tiles. Ben's perimeter is 16 cm longer than Mei's. What are the two rectangles?

X8. A garden is 10 m long and 8 m wide. Two paths 1 m wide cross it: one goes along its length and one goes across its width. What area of the garden is NOT path?

X9. A rectangle 9 cm long and 4 cm wide is cut into 2 pieces. The pieces fit together exactly to make a square. How long is a side of the square? Did the perimeter go up or down?

Lesson 8: Symmetry and Nets

Symmetry and Nets • Week 4

CORE

C1. How many faces, edges and corners does a cube have?

C2. How many lines of symmetry does a square have?

C3. How many lines of symmetry does a rectangle that is not a square have?

C4. On a dice, opposite faces add up to 7. Which number is opposite 2?

C5. How many squares are there in a net of a cube?

C6. Does a row of 6 squares fold into a cube? Explain.

THINK HARDER

T1. A cuboid is 3 cm long, 2 cm wide and 1 cm high. Its net is drawn on 1 cm squared paper. How many squares does the net cover?

T2. Ben puts a dice on the table. The top face shows 3 and the front face shows 5. What numbers are on the bottom face and on the back face?

T3. A dice is on a table. Mei walks round it and can see the top and the four side faces. The numbers she can see add up to 19. What number is on the bottom face?

T4. There are 5 different shapes made of 4 squares joined edge to edge (a line, a square, an L, a T and a zigzag). Which one has exactly 1 line of symmetry?

T5. Can a 2 by 3 rectangle of squares fold into a cube? Explain.

T6. Which shape made of 5 squares joined edge to edge has exactly 2 lines of symmetry?

T7. Ben folds a net into a cube. Each face is a different colour. How many faces can touch the red face along an edge? Could 5 faces touch it?

T8. How many lines of symmetry does a 2 by 4 rectangle of squares have? A 4 by 4 square?

CHALLENGE (non-routine: justify, find every way)

X1. Two dice are glued together face to face, and the two glued faces show the same number. The numbers on all the faces you can still see add up to 36. What number is on the glued faces?

X2. How many different nets of a cube are there? (Turned or flipped copies count as the same net.)

X3. An open box (a cube with no lid) is made from 5 squares. Of the 12 different shapes made of 5 squares joined edge to edge, how many fold into an open box? (Turned or flipped copies count as the same.)

X4. Ravi shades the 4 squares along one diagonal of a 4 by 4 grid (top left to bottom right). How many lines of symmetry does the pattern have?

X5. Ali says: "On a dice, the numbers on the 4 side faces always add up to 14." Is he right? Explain.

X6. A cube has faces numbered 1 to 6 (not a normal dice). Face 1 touches faces 2, 3, 4 and 5. Face 2 touches faces 1, 3, 5 and 6. Which face is opposite 1? Opposite 2? Opposite 3?

X7. Three dice are stacked in a tower on a table. The top face of the tower shows 5. What is the total of the numbers on all the faces you cannot see (the faces touching each other and the table)?

X8. In how many ways can you shade exactly 2 squares of a 3 by 3 grid so that the up-down line through the middle is a line of symmetry?

X9. Shade exactly 5 squares of a 3 by 3 grid so that the pattern has 4 lines of symmetry. There are 2 ways. Find both.

Lesson 9: Data Detectives

Tables and line graphs • Week 5

CORE

C1. On a line graph, each line on the scale is worth 10. A point is halfway between the 30 line and the 40 line. What number does it show?

C2. A stall sold 45 bowls of laksa on Monday and 60 bowls on Tuesday. By how many bowls did sales go up?

C3. A table shows 32 visitors on Monday, 48 on Tuesday and 20 on Wednesday. How many visitors came altogether?

C4. A line graph goes up from 8 am to 10 am, then down until 12 pm. At what time was the number the highest?

C5. A table shows that Ali has 25 cards and Mei has 18 more cards than Ali. How many cards do they have altogether?

C6. Three days of sales add up to 90. Day 1 is 30 and Day 2 is 25. What is Day 3?

THINK HARDER

T1. A table shows that Class 4A has 28 boys and 6 more girls than boys. Class 4B has 63 pupils, with 5 more girls than boys. Which class has more girls?

T2. On a line graph each line on the scale is worth 20. A point is 3 lines above the 40 line. What number does it show?

T3. A shop sold 300 kites in 4 days. Day 1: 65, Day 2: 80, Day 4: 90. How many kites were sold on Day 3?

T4. Ali's height: age 7, 120 cm; age 8, 127 cm; age 9, 131 cm; age 10, 137 cm. In which year did he grow the most? How much did he grow altogether from age 7 to age 10?

T5. Shop A sold 30, 35, 40 and 45 cakes in weeks 1 to 4. Shop B sold 50, 48, 46 and 44 cakes. In which week did Shop A first sell more than Shop B?

T6. A graph's scale starts at 40. Mei says a point at 60 is "double" a point at 50 because it is twice as high on the graph. Is she right? Explain.

T7. Visitors to a museum: Monday 120, Tuesday 95, Wednesday 140, Thursday 110. Between which two days was the biggest change? Was it an increase or a decrease?

T8. The temperature went up by 2 °C every hour from 8 am to 1 pm. At 1 pm it was 33 °C. What was the temperature at 8 am?

CHALLENGE (non-routine: justify, find every way)

X1. A shop's sales go up by the same amount every week. In week 2 it sold 45 toys and in week 6 it sold 85 toys. How many toys did it sell in week 1? In which week did it first sell more than 100 toys?

X2. A stall sold 250 bowls of noodles from Monday to Friday. Monday 40 and Friday 50. Tuesday and Wednesday were the same, and Thursday was 10 more than Tuesday. Find the sales for Tuesday, Wednesday and Thursday.

X3. On Ben's graph each line is worth 2. On Mei's graph each line is worth 10. Both lines go up by 3 grid lines from Monday to Tuesday. Whose number went up more, and by how much more?

X4. Three classes read 210 books altogether in Term 1. In Term 2, each class read 15 more books than in Term 1. How many books did the three classes read in Term 2?

X5. At a pool, the number of swimmers went up by 15 every hour from 9 am to 11 am, then went down by 10 every hour until 1 pm. At 1 pm there were 50 swimmers. How many were there at 9 am?

X6. A stall sold 400 plates of nasi lemak from Monday to Friday. Each day it sold 10 more plates than the day before. How many plates did it sell on Monday?

X7. Four teams scored 100 points altogether. Team A scored twice as many as Team B. Team C scored 10 more than Team B. Team D scored 22. How many points did each team score?

X8. Tank A has 80 litres of water and loses 6 litres every hour. Tank B has 20 litres and gains 9 litres every hour. On a line graph the two lines cross. After how many hours do the tanks hold the same amount? How much is in each?

X9. A graph's scale starts at 200 and each line is worth 50. Monday's point is 2 lines up and Tuesday's point is 6 lines up. Ravi says Tuesday's number is 3 times Monday's. Is he right? Explain.

Lesson 10: Challenge Day

Mixed heuristics • Week 5

CORE

C1. Two next-door numbers (like 7 and 8) multiply to make 90. What are they?

C2. A farm has chickens and rabbits. There are 8 heads and 22 legs. How many rabbits are there?

C3. I think of a number. I add 15, then double it. I get 70. What is my number?

C4. Ali is 6th from the front and 6th from the back of a queue. How many people are in the queue?

C5. Two numbers add up to 30. One is 6 more than the other. What are they?

C6. Ben has 6 coins. Some are 10¢ and some are 50¢. They make $1.40 altogether. How many 50¢ coins does he have?

THINK HARDER

T1. There are 20 bicycles and tricycles with 47 wheels altogether. How many tricycles are there?

T2. Two next-door even numbers (like 6 and 8) multiply to make 624. What are they?

T3. Ravi spent $15 on a book. Then he spent half of the money left on a bag. He had $18 left. How much money did he have at first?

T4. Ali, Ben and Siti are 9, 10 and 11 years old. Ben is older than Siti. Siti is not the youngest. How old is each child?

T5. Mei is 9th from the front and 14th from the back of a queue. How many people are in the queue?

T6. There are 12 spiders (8 legs) and ants (6 legs) with 82 legs altogether. How many spiders are there?

T7. Three next-door numbers add up to 117. What is the biggest of the three?

T8. Siti and her dad are 48 years old altogether. In 6 years, her dad will be 4 times as old as Siti. How old is Siti now?

CHALLENGE (non-routine: justify, find every way)

X1. How many whole numbers from 1 to 100 have the digit 7 in them at least once?

X2. A farm has 20 chickens and goats. The chickens have 4 more legs altogether than the goats. How many chickens are there?

X3. I am a 2-digit number. My digits add up to 11. If you swap my digits, the new number is 27 more than me. Who am I?

X4. Ali, Ben, Mei and Siti sit in a row of 4 seats, numbered 1 to 4 from the left. Mei sits at one end. Ben sits next to Mei. Ali does not sit next to Ben. Ali sits in seat 1 or seat 2. Who sits in each seat?

X5. A cake costs $3 more than a drink. 4 cakes and 3 drinks cost $40. How much does a drink cost?

X6. I halve a number, then add 6, then multiply by 3. I get 48. What was the number?

X7. A hawker centre has 15 tables. Some seat 4 people and some seat 6 people. There are 76 seats altogether. How many tables seat 6?

X8. Mei gave Ali half of her stickers and 2 more. Then she gave Ben half of the stickers left and 2 more. She had 5 stickers left. How many did she have at first?

X9. Ben's uncle is 4 times as old as Ben. In 10 years, his uncle will be only twice as old as Ben. How old is Ben now?

Answer key (tutor only)

Lesson 1: Factor Hunters

C1. 1, 2, 3, 4, 6, 12

C2. 7, 14, 21, 28, 35

C3. Yes: 4 × 13 = 52.

C4. 15

C5. 1 × 20, 2 × 10, 4 × 5

C6. 12

T1. 12 factors: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72.

T2. 18 seconds (6, 12, 18 and 9, 18).

T3. 1, 2, 3, 4, 6, 12

T4. 56 (the multiples of 7 are 56, 63, 70, 77; only 5 + 6 = 11).

T5. 2, 4, 7 or 14 friends.

T6. 24, with 8 factors: 1, 2, 3, 4, 6, 8, 12, 24.

T7. 72 and 108

T8. 12 packs, each with 3 pencils and 4 erasers.

X1. 33. Multiples of 4: 25. Multiples of 6: 16. Both (multiples of 12): 8, counted twice. 25 + 16 − 8 = 33.

X2. 41. One less than the number of eggs is a common multiple of 4 and 5 (20, 40, 60), so 21, 41 or 61; only 41 is between 30 and 50.

X3. 12 (1, 2, 3, 4, 6, 12). Every number below 12 has fewer than 6 factors.

X4. 4: 24, 48, 72, 96.

X5. After 12 minutes: Ravi has run 3 laps and Siti 2 laps.

X6. 21 (1 × 21, 3 × 7), 27 (1 × 27, 3 × 9), 33 (1 × 33, 3 × 11), 35 (1 × 35, 5 × 7) or 39 (1 × 39, 3 × 13). Any one of these.

X7. 10:00 am. The smallest common multiple of 4, 6 and 10 is 60 minutes.

X8. 61. One less than the number is a common multiple of 2, 3, 4, 5 and 6: 60. 60 + 1 = 61 (the next one, 121, is too big).

X9. 1, 4, 9 and 16. Each locker is changed once for every factor of its number. Only numbers with an odd number of factors (1, 4, 9, 16: a number times itself) end up open.

Lesson 2: Big Number Puzzles

C1. 4000 (4 thousands)

C2. 53 080

C3. 28 000

C4. 69 230

C5. 45 099, 45 900, 45 909, 45 990

C6. 6400

T1. 55 555 (66 666 rounds to 70 000; 44 444 rounds to 40 000).

T2. 69 741 (259 away; the closest number above 70 000 is 71 469, which is 1469 away).

T3. Add 2250 each time: 24 000, 26 250.

T4. 100 (from 3450 to 3549).

T5. 79 500

T6. 8 × 6000 = 48 000. More, because 6102 was rounded down (exact: 48 816).

T7. 9 (11 111, 22 222, ... 99 999).

T8. 98 765

X1. 99 456. He says 97 numbers (3456, 4456, ... 99 456).

X2. 500. To the nearest thousand: 44 500 to 45 499. To the nearest ten thousand: 45 000 to 54 999. Both: 45 000 to 45 499.

X3. 100. They look like 4 a b a 4: 10 choices for a and 10 for b.

X4. She rounded to 4000, so the number is from 3500 to 4499.

X5. 10: 49 594, 49 694, 49 794, 49 894, 49 994, 50 005, 50 105, 50 205, 50 305, 50 405.

X6. 56. 99 990 to 99 999 is 10 numbers × 5 digits = 50, and 100 000 has 6 digits.

X7. 15. Ten thousands digit 1: the other 2 goes as 2 in one place (4 ways) or 1 and 1 in two places (6 ways): 10. Ten thousands digit 2: 4 ways. Ten thousands digit 3: 1 way (30 000). 10 + 4 + 1 = 15.

X8. Largest 43 210, smallest 10 234 (the only five different digits that add up to 10 are 0, 1, 2, 3, 4).

X9. 40 004, which is 11 km more.

Lesson 3: Operation Detectives

C1. 36

C2. 72

C3. 16

C4. 84

C5. 7 035

C6. 706

T1. 468 (5 units = 2 340, 1 unit = 468).

T2. (3 + 5) × 6 − 2 = 8 × 6 − 2 = 46.

T3. 12 ÷ 4 × 5 = 15 (the only way).

T4. Multiplying by 5 and then dividing by 5 gets back to your number. The 35 divided by 5 adds 7. Taking away your number leaves the 7. So the answer is always 7.

T5. 20 (18 + 7 = 25, 4 × 25 = 100, 120 − 100 = 20).

T6. 3, 7, 10, 17, 27.

T7. 3 × 4 + 2 × 2 = 16, so $16.

T8. No. 50 − 20 − 10 = 20 (left to right), but 50 − (20 − 10) = 50 − 10 = 40.

T9. 55 (425 + 15 = 440, 440 ÷ 8 = 55).

X1. For example (4 × 4 + 4) ÷ 4 = 20 ÷ 4 = 5.

X2. 16 steps: 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1.

X3. 200 and 800 (5 units = 1 000, 1 unit = 200).

X4. $122 (300 − 56 = 244, 244 ÷ 2 = 122; Ali has $178).

X5. For example 6 × (5 − 4 + 3) = 24 and (3 + 5 − 4) × 6 = 24. Other ways are possible.

X6. 6 (6 × 6 − 6 = 36 − 6 = 30). Try 5: 20, too small; try 7: 42, too big.

X7. 2 ways: 1 − 2 + 3 + 4 − 5 + 6 = 7 and 1 + 2 − 3 − 4 + 5 + 6 = 7. (All + makes 21, which is 14 too many. Changing a + to a − takes that number away twice, so the numbers we change must add up to 7: 2 and 5, or 3 and 4.)

X8. $48. Work backwards: $9 was half of what was left, so before the gift she had $18. Before lunch: $18 + $6 = $24. That was half, so at first she had $48.

X9. 3. Work backwards: 4th = 49 − 30 = 19, 3rd = 30 − 19 = 11, 2nd = 19 − 11 = 8, 1st = 11 − 8 = 3. The chain is 3, 8, 11, 19, 30, 49.

Lesson 4: Fraction Frontier

C1. 3 2/3

C2. 11/4

C3. 8

C4. 12

C5. 5/6 (2/3 = 4/6)

C6. 7/10 (2/5 = 4/10)

T1. Ben. 7/4 = 14/8 = 1 6/8, which is more than 1 5/8.

T2. 25 (1 unit = 5, blue = 15, not blue = 5 units = 25).

T3. 1/2, 7/12, 2/3, 5/6 (6/12, 7/12, 8/12, 10/12).

T4. 17/6 = 2 5/6. Between 2 and 3: 13/6, 14/6, 15/6, 16/6, 17/6. Odd numerators: 13/6, 15/6, 17/6. 15/6 = 2 1/2 is exactly in the middle and 13/6 is closer to 2, so only 17/6 is closer to 3.

T5. 3/10 and 4/10 (1/5 = 2/10 and 1/2 = 5/10 do not count).

T6. 12 pieces (4 in each metre).

T7. 19 (4 wholes = 16 quarters, plus 3).

T8. No. Both have 3 pieces, but sevenths are smaller pieces than fifths, so 3/5 is bigger.
T9. 40 (2 units = 16, 1 unit = 8, 5 units = 40).
X1. 4: 4/12, 5/12, 6/12, 7/12 (1/4 = 3/12 and 2/3 = 8/12 do not count).
X2. $60. Book 4/12, pen 3/12, so 5/12 is left. 5 units = $25, 1 unit = $5, 12 units = $60.
X3. 40 litres. 3/4 = 6/8, so 15 litres is 3/8. 1/8 = 5 litres, 8/8 = 40 litres.
X4. 5/7. It is 2/7 short of 1. The next best, 2/3, is 1/3 = 2/6 short, and 2/7 is less than 2/6.
X5. 24. Red 3/12 and blue 4/12, so green is 5/12. 5 units = 10, 1 unit = 2, 12 units = 24.
X6. 5 pieces (Mei 3 pieces, Ben 4 pieces, 12 − 7 = 5).
X7. 12/16. 3/4 has 3 + 4 = 7 parts; 28 ÷ 7 = 4, so 3 × 4 = 12 and 4 × 4 = 16.
X8. 18. Half = 3 sixths and a third = 2 sixths, so 1 sixth = 3 and the number is 6 sixths = 18. Check: 9 is 3 more than 6.
X9. 150 pages. Cut each fifth in half to make tenths: Monday 4 tenths, the rest is 6 tenths, Tuesday 3 tenths, 3 tenths left = 45. 1 tenth = 15, 10 tenths = 150.

Lesson 5: Decimal Detectives

C1. 3.407
C2. 6 hundredths (0.06)
C3. 0.9. Line up the points: 0.90 and 0.89. 9 tenths is more than 8 tenths.
C4. 7.5
C5. 16.0
C6. $3.75
T1. 0, 1, 2 and 3 (3.45 is more than 3.42)
T2. Smallest 9.50, largest 10.49
T3. 2.65 and 2.605
T4. 0.8 m each; 3.2 m
T5. $1.85 ($10 − $2.60 = $7.40, $7.40 ÷ 4 = $1.85)
T6. 0.06, 0.066, 0.6, 0.606, 0.66 (as 0.060, 0.066, 0.600, 0.606, 0.660)
T7. 1.44 m (Ben 1.53 m)
T8. 47.2 (2.8 away; the next closest is 42.7)
X1. 2.459 (it is from 2.455 to 2.464; 2.45_ needs 9 to make 20; 2.46_ would need 8, but 2.468 rounds to 2.47)
X2. 0.2 kg (2 apples = 0.95 − 0.45 = 0.5 kg, 1 apple = 0.25 kg, pear = 0.45 − 0.25)
X3. Six 50¢ and four 20¢ (ten 20¢ make $2; each swap to a 50¢ adds 30¢; $1.80 more needs 6 swaps)
X4. 5.50, 5.51, 5.52, 5.53, 5.54
X5. $1.00 (3 pencils + $1.70 = $4.70, so 3 pencils cost $3.00; a pen costs $1.85)
X6. 19 (3.91 to 3.99 is 9, 4.00 to 4.09 is 10)
X7. 48.21 (1.79 away; 48.12 is 1.88 away, and numbers starting with 8 are far too big)
X8. Ali $2.30, Ben $2.80, Siti $4.10 (take away $0.50 + $1.80 = $2.30, then $6.90 ÷ 3 = $2.30)
X9. 0.25 kg (half the water is 1.45 − 0.85 = 0.6 kg, all the water is 1.2 kg, jug = 1.45 − 1.2)

Lesson 6: Angle Turns

C1. 180°
C2. 270°
C3. S
C4. SW
C5. 180°
C6. 26°
T1. N (225° is 5 steps of 45°: S, SW, W, NW, N)
T2. 60° and 60°. Both are 2 spaces apart the shorter way.
T3. 210° (35 minutes × 6°)
T4. Two ways: 45° clockwise and 135° anticlockwise; or 90° clockwise and 180°.
T5. 225° (W → NW → N → NE → E → SE: 5 steps of 45°)
T6. 6 angles (25°, 40°, 25°, 65°, 65°, 90°); the biggest is 90°.
T7. 270° (180° + 90°)

T8. 63° (90° – 27°)

T9. 18° and 72° (5 equal parts make 90°, so one part is 18°)

X1. 75° (the minute hand points to 6; the hour hand is halfway from 3 to 4, so it is 2 and a half spaces from the 6: 30° + 30° + 15°)

X2. 10 angles (4 + 3 + 2 + 1). Each small angle is 30°, so the right angles are the 3-part angles: 2 of them.

X3. 8 turns. After 30 turns it faces W (30 = 3 × 8 + 6, and 6 steps from N is W).

X4. 15 minutes; 3 hours

X5. 10° (it turns 30° in 60 minutes)

X6. 90° (Mei 225°, Ali 135°)

X7. 105° (the hour hand is halfway between 9 and 10; from it to the 6 is 3 and a half spaces: 3 × 30° + 15°)

X8. 15°, 30° and 45° (6 equal parts make 90°, so one part is 15°)

X9. 20° (90° + 90° = 180°, and 180° – 160° = 20°)

Lesson 7: Area Architects II

C1. 63 cm²

C2. 38 cm

C3. 81 cm² (each side is 9 cm).

C4. 30 cm (the breadth is 7 cm; 8 + 7 + 8 + 7 = 30).

C5. 24 + 6 = 30 cm²

C6. 100 – 16 = 84 cm². The perimeter has not changed: it is still 40 cm.

T1. 1 by 30 (62 cm), 2 by 15 (34 cm), 3 by 10 (26 cm), 5 by 6 (22 cm). Smallest: 5 cm by 6 cm, 22 cm.

T2. 1 by 9 (9 cm²), 2 by 8 (16 cm²), 3 by 7 (21 cm²), 4 by 6 (24 cm²), 5 by 5 (25 cm²). Largest: the 5 cm by 5 cm square, 25 cm².

T3. Area 96 – 4 × 9 = 60 cm². Perimeter 40 cm, the same as the rectangle: each cut takes away 3 + 3 and adds 3 + 3.

T4. Area 4 + 9 + 16 = 29 cm². Perimeter 26 cm: push the steps out to a 9 cm by 4 cm rectangle.

T5. 80 m². The whole thing is 14 m by 10 m = 140 m²; take away the lawn 60 m².

T6. Smallest 12 cm (a 2 by 4 rectangle, or a 3 by 3 square with one corner missing). Largest 18 cm (for example a row of 8).

T7. 8 cm longer: 28 cm instead of 20 cm. The area stays 24 cm².

T8. 8 cm by 5 cm. (List the pairs for 40: 1 by 40, 2 by 20, 4 by 10, 5 by 8; only 5 by 8 has perimeter 26 cm.)

X1. 64 cm². Each rectangle is one whole side and one half side all round, which is 3 sides of the square. 24 ÷ 3 = 8 cm, and 8 × 8 = 64.

X2. Area 60 – 8 = 52 cm². Perimeter 32 + 4 + 4 = 40 cm: the two 4 cm sides inside the gap cannot be pushed out.

X3. 36 m², a 6 m by 6 m square.

X4. 72 m²: the two sides from the wall are 6 m and the side opposite the wall is 12 m (6 + 12 + 6 = 24).

X5. Area 16 + 16 – 4 = 28 cm² (the 2 by 2 middle is counted once). Perimeter 32 cm: each of the 4 arms is 3 cm long, so each arm adds 3 + 2 + 3 = 8 cm.

X6. 32 m². The sides go from 7 m to 9 m, and 81 – 49 = 32.

X7. Mei 6 cm by 6 cm (24 cm), Ben 2 cm by 18 cm (40 cm). The perimeters for 36 are 74, 40, 30, 26 and 24; only 40 and 24 differ by 16.

X8. 63 m². Slide the paths to the edges: the grass makes a 9 m by 7 m rectangle. (Or 80 – 10 – 8 + 1 = 63, because the square where the paths cross was counted twice.)

X9. Area 36 cm², so the side is 6 cm. The perimeter went down from 26 cm to 24 cm.

Lesson 8: Symmetry and Nets

C1. 6 faces, 12 edges, 8 corners.

C2. 4

C3. 2 (the diagonals are not lines of symmetry).

C4. 5

C5. 6

C6. No. Four squares wrap round to make the walls; the 5th and 6th land on top of the first two, and the two ends are left open.

T1. 22 squares: 2 faces of 3 by 2 (6 each), 2 faces of 3 by 1 (3 each) and 2 faces of 2 by 1 (2 each). 12 + 6 + 4 = 22.

T2. Bottom 4 (3 + 4 = 7). Back 2 (5 + 2 = 7).

T3. 2. All six faces add up to 1 + 2 + 3 + 4 + 5 + 6 = 21, and 21 − 19 = 2.

T4. Only the T. The line has 2, the square has 4, the L and the zigzag have 0.

T5. No. It has a 2 by 2 block of squares: 4 squares meet at one corner, but on a cube only 3 faces meet at a corner, so two of them would overlap.

T6. Only the row of 5 squares (one line up-down through the middle square and one line along the row).

T7. 4 faces touch it. 5 cannot: the face opposite red never touches it.

T8. 2 and 4.

X1. 3. Two dice have 21 + 21 = 42 altogether. The hidden faces add up to 42 − 36 = 6, and they show the same number, so each is 3.

X2. 11.

X3. 8. Only 4 shapes do not fold: the row of 5, the V (an L with arms of 3 and 3), the U (a C shape) and the P (a 2 by 2 block with one extra square).

X4. 2: the diagonal it lies on, and the other diagonal of the grid.

X5. Yes. The 4 side faces are two opposite pairs, and each pair adds up to 7: 7 + 7 = 14.

X6. Opposite 1 is 6 (the only face it does not touch). Opposite 2 is 4. That leaves 3 and 5, so opposite 3 is 5.

X7. 16. The top dice: its bottom is 2 (5 + 2 = 7). The middle dice: top and bottom are opposite, 7. The bottom dice: top and bottom, 7. 2 + 7 + 7 = 16.

X8. 6. Both squares in the middle column: 3 ways. One square in each side column, in the same row: 3 ways.

X9. The middle square and the 4 corners (an X), or the middle square and the 4 edge-middle squares (a plus).

Lesson 9: Data Detectives

C1. 35

C2. 15 bowls

C3. 100

C4. 10 am

C5. 68 (Mei has 43; 25 + 43 = 68)

C6. 35

T1. Neither: both have 34 girls. (4A: 28 + 6 = 34. 4B: 63 − 5 = 58, 58 ÷ 2 = 29 boys, 29 + 5 = 34 girls.)

T2. 100 (40 + 20 + 20 + 20)

T3. 65 (300 − 65 − 80 − 90)

T4. From age 7 to 8 (7 cm). Altogether 17 cm.

T5. Week 4 (45 is more than 44).

T6. No. 60 is only 10 more than 50. It looks twice as high because 50 is 10 above the bottom (40) and 60 is 20 above it.

T7. Tuesday to Wednesday: an increase of 45. (The other changes are down 25 and down 30.)

T8. 23 °C (5 hours, so 10 °C less)

X1. Week 1: 35 (4 jumps make 40, so 10 a week). First more than 100: week 8 (105).

X2. Tuesday 50, Wednesday 50, Thursday 60. (Tue + Wed + Thu = 160; take away the extra 10, 150 shared by 3 = 50.)

X3. Mei's: hers went up 30 and Ben's went up 6, so 24 more.

X4. 255 (210 + 15 + 15 + 15)

X5. 40. Work backwards: 12 pm 60, 11 am 70, 10 am 55, 9 am 40.

X6. 60 (60, 70, 80, 90, 100)

X7. A 34, B 17, C 27, D 22. (A + B + C = 78: four equal parts and 10 more, so one part is 68 ÷ 4 = 17.)

X8. After 4 hours: 56 litres each. The gap of 60 litres closes by 15 litres every hour.

X9. No. Monday is 300 and Tuesday is 500. 3 times 300 is 900.

Lesson 10: Challenge Day

C1. 9 and 10

C2. 3 rabbits (and 5 chickens). Suppose all chickens: 16 legs, 6 short, each swap adds 2.

C3. 20 (70 ÷ 2 = 35, 35 − 15 = 20)

C4. 11 (5 + 1 + 5)

C5. 12 and 18

C6. 2 (and four 10¢ coins: 100¢ + 40¢ = 140¢)

T1. 7 tricycles (and 13 bicycles). All bicycles: 40 wheels, 7 short, each swap adds 1.

T2. 24 and 26

T3. $51 (18 × 2 = 36, 36 + 15 = 51)

T4. Ben 11, Siti 10, Ali 9.

T5. 22 (8 + 1 + 13)

T6. 5 spiders (and 7 ants). All ants: 72 legs, 10 short, each swap adds 2.

T7. 40 (38 + 39 + 40)

T8. 6 (in 6 years the total is 60: 5 equal bars, so Siti will be 12)

X1. 19 (7, 17, 27, 37, 47, 57, 67, 87, 97 and 70 to 79; 77 is counted once)

X2. 14 chickens (28 legs) and 6 goats (24 legs).

X3. 47 (74 – 47 = 27)

X4. Seat 1 Ali, seat 2 Siti, seat 3 Ben, seat 4 Mei.

X5. $4 (a cake is $7). Suppose all 7 are drinks: take away 4 × $3 = $12, then $28 ÷ 7 = $4.

X6. 20 (48 ÷ 3 = 16, 16 – 6 = 10, 10 × 2 = 20)

X7. 8 tables seat 6 (and 7 seat 4). All seat 4: 60 seats, 16 short, each swap adds 2.

X8. 32. Work backwards: (5 + 2) × 2 = 14, (14 + 2) × 2 = 32.

X9. 5 (uncle 20; in 10 years 15 and 30)